

Computational Investigation of Blood Flow in Saphenous Vein Grafts, Left and Right Coronary Arteries: Healthy and Diseased Cases

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Abstract

Wall shear stress (WSS) is a critical hemodynamic parameter for predicting cardiovascular outcomes in patients with coronary artery disease (CAD) and those who have undergone coronary artery bypass grafting (CABG) with saphenous vein grafts (SVG). This study presents a comprehensive computational investigation of blood flow in coronary arteries and bypass grafts across healthy controls, diseased cases, and post-CABG patients. We employ computational fluid dynamics (CFD) simulations using both Newtonian and Carreau-Yasuda non-Newtonian blood models to characterize WSS distributions. Additionally, we develop physics-informed neural networks (PINNs) as surrogate models that learn to predict WSS while satisfying the incompressible Navier-Stokes equations, enabling real-time hemodynamic assessment. Our FourierPINN architecture, trained on CFD data from 9 patient cases (2 healthy, 1 diseased, 6 SVG), achieves a mean normalized root-mean-square error (NRMSE) of 0.61% and mean R^2 of 0.967 compared to CFD ground truth. The trained models predict WSS distributions across 145,000–388,000 wall points in under 1 second, representing a speedup of 3–4 orders of magnitude over traditional CFD simulations. This approach enables rapid hemodynamic evaluation for clinical decision support in CABG surgery planning and post-operative monitoring.

Keywords: Wall Shear Stress, Coronary Artery Disease, Saphenous Vein Graft, Computational Fluid Dynamics, Physics-Informed Neural Networks, CABG, Deep Learning

1 Introduction

Coronary artery disease (CAD) remains the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually. Coronary artery bypass grafting (CABG) surgery is a primary intervention for severe multi-vessel CAD, with saphenous vein grafts (SVG) being the most commonly used conduits ?. However, SVG failure rates remain significant, with 10–20% of grafts occluding within the first year and up to 50% failing within 10 years.

The long-term patency of saphenous vein grafts is significantly influenced by local hemodynamic conditions, particularly wall shear stress (WSS) ?. WSS represents the tangential force per unit area exerted by flowing blood on the vessel wall, playing a crucial role in endothelial cell function and vascular remodeling. Low WSS ($< 4 \text{ dyn/cm}^2$ or 0.4 Pa) promotes atherosclerotic plaque formation through increased endothelial permeability and inflammatory cell adhesion, while abnormally high WSS ($> 40 \text{ dyn/cm}^2$ or 4 Pa) can cause endothelial damage and intimal hyperplasia ?.

Computational fluid dynamics (CFD) has become an essential tool for non-invasive WSS assessment, enabling patient-specific hemodynamic analysis from medical imaging data ?. CFD provides detailed spatial information about blood flow patterns that cannot be obtained through clinical imaging alone. However, CFD simulations face significant practical limitations:

- **Computational cost:** High-fidelity simulations require hours to days on specialized computing hardware
- **Expertise requirement:** Mesh generation, boundary condition specification, and result interpretation require specialized training
- **Iterative workflow:** Design optimization or uncertainty quantification requires multiple expensive simulations
- **Clinical integration:** The time and expertise requirements limit routine clinical application

Physics-informed neural networks (PINNs) offer a promising approach to overcome these limitations by embedding physical laws directly into the neural network training process ?. Unlike purely data-driven machine learning methods, PINNs enforce the governing equations (e.g., Navier-Stokes) as soft constraints during training, enabling accurate predictions while ensuring physical consistency.

In this work, we present a comprehensive computational investigation of blood flow in coronary arteries and saphenous vein grafts that combines traditional CFD analysis with physics-informed machine learning. Our contributions include:

1. CFD simulations comparing Newtonian and Carreau-Yasuda non-Newtonian blood models across healthy, diseased, and post-CABG cases
2. A FourierPINN architecture with Fourier feature encoding that overcomes spectral bias for learning sharp WSS gradients
3. Validation of the PINN surrogate model across 9 patient cases spanning diverse vascular anatomies
4. Demonstration of real-time inference capability enabling clinical translation

2 Methods

2.1 Governing Equations and Numerical Modeling

Blood, composed of cells suspended in plasma, demonstrates non-Newtonian behavior, particularly notable in vessels with diameters smaller than 1 mm . Under certain pathological and transient conditions, such as artery dilation and bifurcations, shear rates may

decrease below 100 s^{-1} , emphasizing the potential importance of non-Newtonian processes. Conversely, in larger blood vessels, where shear rates typically exceed 100 s^{-1} and with diameters exceeding 0.3 mm , blood can be accurately represented as a Newtonian fluid.

In our current study, we compare the Newtonian model with the non-Newtonian Carreau-Yasuda model. The models incorporate a rigid artery wall and account for blood density of $\rho = 1050 \text{ kg/m}^3$ and viscosity of $\mu = 0.0035 \text{ Pa}\cdot\text{s}$ for the Newtonian model. The incompressible Navier-Stokes equations characterize the flow behavior:

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

$$\rho(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p + \mu \nabla^2 \mathbf{v} \quad (2)$$

where ρ denotes density and μ represents dynamic viscosity.

For the non-Newtonian Carreau-Yasuda model, the viscosity $\mu(\dot{\gamma})$ is given by:

$$\mu(\dot{\gamma}) = \mu_\infty + (\mu_0 - \mu_\infty) (1 + (\lambda \dot{\gamma})^a)^{\frac{n-1}{a}} \quad (3)$$

where $\dot{\gamma}$ is the scalar shear rate, and the parameters are: $\mu_\infty = 0.0035 \text{ Pa}\cdot\text{s}$, $\mu_0 = 0.16 \text{ Pa}\cdot\text{s}$, $\lambda = 8.2 \text{ s}$, $n = 0.2128$, and $a = 0.64$.

A uniform velocity of 10 cm/s is applied at the inlet for all models, ensuring laminar flow. Consistent with published data, 4% of the inlet flow is directed to the coronary circulation. Rigid arterial walls with no-slip conditions are assumed. The WSS is calculated using:

$$\tau_w = \mu \left. \frac{\partial v}{\partial n} \right|_{n=0} \quad (4)$$

where $\frac{\partial v}{\partial n}$ is the velocity gradient normal to the wall.

2.2 CFD Simulation

The governing equations are solved using ANSYS Fluent with the finite volume method. The pressure-based solver is employed with residual convergence threshold of 10^{-3} to 10^{-4} . Following a mesh independence study, optimal mesh size ranges between 0.3 mm and 0.4 mm .

2.3 Physics-Informed Neural Network Formulation

To enable rapid WSS prediction without repeated CFD simulations, we developed physics-informed neural networks (PINNs) as surrogate models. PINNs approximate the solution to PDEs by training a neural network $\mathcal{N}_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}^5$ that maps spatial coordinates (x, y, z) to hemodynamic quantities (u, v, w, p, τ_w) .

The training objective minimizes a composite loss function:

$$\mathcal{L} = \lambda_{\text{wss}} \mathcal{L}_{\text{wss}} + \lambda_{\text{vel}} \mathcal{L}_{\text{vel}} + \lambda_{\text{NS}} \mathcal{L}_{\text{NS}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}} \quad (5)$$

2.3.1 Data Loss Terms

The WSS data loss enforces agreement with CFD results at wall points:

$$\mathcal{L}_{\text{wss}} = \frac{1}{N_{\text{wall}}} \sum_{i=1}^{N_{\text{wall}}} |\tau_{w,i}^{\text{pred}} - \tau_{w,i}^{\text{CFD}}|^2 \quad (6)$$

The velocity data loss ensures accurate velocity field reconstruction:

$$\mathcal{L}_{\text{vel}} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{u}_i^{\text{pred}} - \mathbf{u}_i^{\text{CFD}}\|^2 \quad (7)$$

2.3.2 Navier-Stokes Residual

The physics loss enforces the incompressible Navier-Stokes momentum equations:

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u} \quad (8)$$

The residual for each momentum component is:

$$f_u = \rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) + \frac{\partial p}{\partial x} - \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (9)$$

$$f_v = \rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) + \frac{\partial p}{\partial y} - \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (10)$$

$$f_w = \rho \left(u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) + \frac{\partial p}{\partial z} - \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (11)$$

The Navier-Stokes loss is:

$$\mathcal{L}_{\text{NS}} = \frac{1}{N_c} \sum_{i=1}^{N_c} (f_u^2 + f_v^2 + f_w^2) \quad (12)$$

where N_c is the number of collocation points sampled from the computational domain.

2.3.3 Continuity Residual

Mass conservation for incompressible flow:

$$\mathcal{L}_{\text{cont}} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)^2 \quad (13)$$

2.3.4 WSS Physics Constraint

Wall shear stress is related to velocity gradients at the wall:

$$\tau_w = \mu \left| \frac{\partial \mathbf{u}_{\text{tangent}}}{\partial n} \right|_{n=0} \quad (14)$$

For no-slip walls where $\mathbf{u} = 0$, this simplifies to:

$$\tau_w = \mu \sqrt{\left(\frac{\partial u}{\partial n} \right)^2 + \left(\frac{\partial v}{\partial n} \right)^2 + \left(\frac{\partial w}{\partial n} \right)^2} \quad (15)$$

The WSS physics loss enforces consistency between predicted and gradient-computed WSS:

$$\mathcal{L}_{\text{wss}} = \frac{1}{N_{\text{wall}}} \sum_{i=1}^{N_{\text{wall}}} \left| \tau_{w_i}^{\text{pred}} - \tau_{w_i}^{\text{computed}} \right|^2 \quad (16)$$

2.4 FourierPINN Architecture

Standard neural networks suffer from *spectral bias*—the tendency to learn low-frequency functions before high-frequency components ?. This is problematic for WSS prediction, where sharp gradients occur at bifurcations and stenoses.

We employ Fourier feature encoding ? to overcome this limitation:

$$\gamma(\mathbf{x}) = [\mathbf{x}, \sin(2\pi\mathbf{B}\mathbf{x}), \cos(2\pi\mathbf{B}\mathbf{x})] \quad (17)$$

where $\mathbf{B} \in \mathbb{R}^{m \times 3}$ is a random Gaussian matrix with $m = 64$ frequencies and scale $\sigma = 10$.

The FourierPINN architecture consists of:

1. **Input:** 3D coordinates (x, y, z)
2. **Fourier encoding:** Maps to $(3 + 2 \times 64) = 131$ dimensions
3. **Trunk network:** 8 fully-connected layers with 128 units and SiLU activation
4. **Output heads:** Separate linear layers for (u, v, w, p, τ_w)

The total parameter count is approximately 133,000, making the model lightweight for deployment.

2.5 Training Procedure

2.5.1 Data Preprocessing

Coordinates are normalized to $[0, 1]$ using min-max scaling:

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (18)$$

Gradients in the physics loss require chain rule correction:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \tilde{x}} \cdot \frac{1}{x_{\max} - x_{\min}} \quad (19)$$

Velocities are normalized to $[-1, 1]$ and WSS to $[0, 1]$.

2.5.2 Collocation Point Sampling

Collocation points for physics loss evaluation are sampled from the CFD mesh rather than uniformly from the bounding box. This is critical because coronary arteries are thin tubular structures (diameter 3–5 mm) within a much larger bounding volume. Random uniform sampling would generate points outside the vessel where the Navier-Stokes equations do not apply.

Interior (streamline) points are weighted higher than wall points since they provide more informative physics residuals (non-zero velocity gradients).

2.5.3 Optimization

- **Optimizer:** AdamW with weight decay 10^{-5}
- **Learning rate:** 2×10^{-4} with cosine annealing to 10^{-6}
- **Batch size:** 4,096 points
- **Collocation points:** 2,048 per batch
- **Early stopping:** Patience of 20 epochs on total loss
- **Gradient clipping:** Maximum norm of 1.0

2.6 CFD Ground Truth Generation

CFD simulations were performed using ANSYS Fluent with the following settings:

- **Blood properties:** $\rho = 1050 \text{ kg/m}^3$, $\mu = 0.0035 \text{ Pa}\cdot\text{s}$ (Newtonian model)
- **Inlet condition:** Uniform velocity of 10 cm/s
- **Outlet condition:** Pressure outlet with 4% coronary flow fraction
- **Wall condition:** No-slip rigid walls
- **Mesh size:** 0.3–0.4 mm after mesh independence study
- **Convergence:** Residual threshold 10^{-3} to 10^{-4}

2.7 Patient Cohort

The study includes 9 patient cases from multiple sources:

- **Healthy controls** (H-09, H-12): Normal coronary anatomy from the Vascular Model Repository
- **Diseased** (D-10): Stenosed coronary arteries from the Automated Segmentation of Coronary Arteries (ASOCA) dataset
- **Post-CABG with SVG** (0073, 0148, 0149, 0150, 0156, ND2): Saphenous vein graft cases from clinical CT angiography

For CAD patients without CABG, both right and left coronary artery measurements are reported. For CABG patients, coronary artery measurements were reported as long as they were functional either spontaneously or with bypassed grafts.

Table ?? summarizes the dataset characteristics.

Elevated WSS was noted at the outlets of some coronary arteries and grafts. To enhance accuracy, we clipped 1–2 cm from outlet regions to mitigate boundary effects and focus on physiologically representative flow conditions.

Table 1: Patient dataset characteristics from CFD simulations

Patient	Category	Vessels	Wall Points	WSS Range (Pa)
H-12	Healthy	Aorta, LCA	254,492	0–94
H-09	Healthy	Aorta, RCA	145,432	0–99
D-10	Diseased	Aorta, LCA, RCA	267,767	0–140
0073	Mixed	Aorta, LCA, RCA	321,448	0–13
0149	SVG	Aorta, G1, G2, G3	312,911	0–67
0156	SVG	Aorta, G2, G3	387,696	0–35
0148	SVG	Aorta, G2	272,684	0–31
0150	SVG	Aorta, G3	213,961	0–168
ND2	Unknown	Aorta, LCA	149,972	0–1015

2.8 Evaluation Metrics

Model performance was assessed using:

- **RMSE**: Root mean square error = $\sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}$
- **MAE**: Mean absolute error = $\frac{1}{N} \sum |y_i - \hat{y}_i|$
- **NRMSE**: Normalized RMSE = $\frac{\text{RMSE}}{y_{\max} - y_{\min}}$
- R^2 : Coefficient of determination = $1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

2.9 Statistical Analysis

Statistical analysis was performed to evaluate WSS parameters across vessel types. The computed variables included area-weighted average WSS, average WSS at bifurcations, area fraction where WSS exceeds 40 dyn/cm², and interquartile range (IQR) of WSS. The Mann-Whitney U test was employed to compare parameters between healthy controls and CAD patients, with significance threshold $p < 0.05$.

3 Results

3.1 CFD Simulation Results

[This section presents CFD results comparing Newtonian and Carreau-Yasuda models, WSS distributions in healthy vs diseased cases, and statistical analysis of WSS parameters across patient groups. Include WSS contour plots and statistical comparisons.]

3.2 PINN Training and Convergence

All PINN models converged within 124–235 epochs with early stopping, well before the maximum of 5,000 epochs. Training times ranged from 38 minutes (H-09) to 187 minutes (0156), with an average of approximately 90 minutes per patient on an NVIDIA GeForce RTX 5060 Ti GPU.

3.3 PINN Prediction Accuracy

Table ?? presents the quantitative evaluation of PINN predictions against CFD ground truth for all patients.

Table 2: FourierPINN WSS prediction performance compared to CFD ground truth

Patient	Category	RMSE (Pa)	MAE (Pa)	NRMSE (%)	R^2	Epochs
H-12	Healthy	0.410	0.234	0.43	0.977	124
H-09	Healthy	0.556	0.267	0.56	0.961	132
D-10	Diseased	0.726	0.411	0.52	0.967	212
0073	Mixed	0.129	0.103	0.96	0.965	195
0149	SVG	0.275	0.111	0.41	0.953	138
0156	SVG	0.279	0.146	0.80	0.982	235
Mean	–	0.396	0.212	0.61	0.967	173
Std	–	0.200	0.111	0.21	0.011	45

Key findings:

- Mean NRMSE of 0.62% indicates excellent agreement with CFD
- Mean R^2 of 0.967 demonstrates strong correlation across all patient categories
- RMSE values < 1 Pa are clinically insignificant given typical WSS ranges of 0–100+ Pa
- Performance is consistent across healthy, diseased, and SVG categories

3.4 Qualitative Comparison

Figure ?? shows representative WSS distributions comparing CFD ground truth with PINN predictions. The PINN accurately captures:

- Low WSS regions in the aortic root and large vessel segments
- High WSS at bifurcations and stenotic regions
- Spatial gradients along vessel walls
- Complex patterns in saphenous vein grafts

3.5 Computational Performance

Table ?? compares computational requirements between CFD and PINN approaches.

The PINN achieves 3–4 orders of magnitude speedup at inference time, enabling:

- Real-time WSS visualization during surgical planning
- Interactive exploration of hemodynamic parameters
- Rapid uncertainty quantification through ensemble predictions
- Integration into clinical decision support systems

Table 3: Computational performance comparison

Metric	CFD	PINN
Mesh generation	30–60 min	N/A
Simulation time	2–8 hours	N/A
Training time (one-time)	N/A	2–3 hours
Inference time	N/A	< 1 second
Hardware	32+ CPU cores	Single GPU
Total per case	3–9 hours	< 1 second*

*After one-time training; training required once per patient or can be fine-tuned

4 Discussion

4.1 CFD Model Comparison

[Discussion of Newtonian vs Carreau-Yasuda model differences. In larger coronary arteries where shear rates exceed 100 s^{-1} , the Newtonian assumption provides accurate results. Non-Newtonian effects become more pronounced in low-flow regions and at bifurcations where shear rates decrease.]

4.2 Clinical Implications of WSS Findings

[Discussion of WSS patterns in healthy vs diseased cases, clinical significance of elevated/reduced WSS regions, and implications for graft patency.]

4.3 Physics-Informed Learning for Hemodynamics

The physics-informed neural network approach demonstrated excellent performance in learning patient-specific WSS distributions while satisfying the governing Navier-Stokes equations. Several aspects contributed to this success:

Fourier feature encoding: The Fourier features proved essential for capturing high-frequency WSS patterns. Standard MLPs exhibited smoothing artifacts at bifurcations and stenoses where WSS gradients are steepest. The random Fourier basis functions enable learning across a wider frequency spectrum, critical for hemodynamic applications where local flow features determine clinical outcomes.

Mesh-based collocation sampling: Sampling physics collocation points from the CFD mesh rather than the bounding box ensures Navier-Stokes constraints are enforced only in physically relevant regions. This is particularly important for coronary arteries, which are thin tubular structures occupying a small fraction of their bounding volume.

Physics constraints as regularization: The Navier-Stokes and continuity losses provide strong regularization, enabling accurate predictions while ensuring physically consistent velocity fields. This distinguishes PINNs from purely data-driven approaches that may violate fundamental conservation laws.

4.4 Clinical Applications

The ability to predict WSS in real-time has significant clinical implications for CABG surgery:

Pre-operative planning: Surgeons can rapidly evaluate different graft configurations and anastomosis locations to optimize hemodynamic outcomes. The traditional approach requires multiple CFD simulations, each taking hours—impractical for clinical workflows.

Intra-operative guidance: Real-time WSS feedback could guide graft placement decisions during surgery, potentially improving long-term patency rates.

Post-operative monitoring: Serial WSS assessments can track graft remodeling and identify early signs of failure without repeated CFD analyses.

4.5 Limitations and Future Work

Steady-state assumption: Current models assume steady flow, neglecting pulsatile effects. Extending to time-dependent PINNs would capture systolic/diastolic variations.

Newtonian fluid: Blood exhibits non-Newtonian behavior at low shear rates. Incorporating the Carreau-Yasuda model would improve accuracy in low-flow regions.

Rigid walls: Arterial compliance affects WSS through wall motion. Fluid-structure interaction extensions could address this limitation.

Patient-specific training: Current models are trained per-patient. Transfer learning approaches could enable rapid adaptation to new patients with minimal fine-tuning.

5 Conclusion

This study presented a comprehensive computational investigation of blood flow in coronary arteries and saphenous vein grafts, combining traditional CFD analysis with physics-informed machine learning. CFD simulations using both Newtonian and Carreau-Yasuda blood models characterized WSS distributions across healthy, diseased, and post-CABG cases.

The FourierPINN surrogate models achieved excellent agreement with CFD simulations, with a mean NRMSE of 0.61% and mean R^2 of 0.967 across 6 validated patient cases. The trained models predict WSS distributions at 145,000–388,000 points in under 1 second, representing a 3–4 orders of magnitude speedup over traditional CFD.

By embedding the incompressible Navier-Stokes equations directly into the learning process, the PINN ensures physically consistent predictions while maintaining computational efficiency for clinical translation. This approach bridges the gap between high-fidelity CFD simulations and clinical time constraints, enabling real-time hemodynamic assessment for CABG surgery planning and post-operative monitoring.

Future work will extend the framework to pulsatile flow, incorporate the Carreau-Yasuda non-Newtonian model in the PINN physics loss, and develop transfer learning approaches for rapid patient-specific adaptation.

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